



LUMS



Evaluation of Classifiers

The Confusion Matrix

Agha Ali Raza



The Output of a Classifier

#	Height (inches)	Weight (kgs)	B.P. Sys	B.P. Dia	Heart disease	
	$\vec{x}$				y	$h(\vec{x})$
1	62	70	120	80	No	No
2	72	90	110	70	No	Yes
3	74	80	130	70	No	No
4	65	120	150	90	Yes	Yes
5	67	100	140	85	Yes	No
6	64	110	130	90	No	Yes
7	69	150	170	100	Yes	Yes
8	75	127	160	95	Yes	No
9	66	66	135	90	Yes	Yes

- y : Gold standards / Gold labels / Ground truth
- $h(\vec{x})$: Predicted labels

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7	69	150	170	100	Yes	Yes
8	75	127	160	95	Yes	No
9	66	66	135	90	Yes	Yes

Negative

Positive

Negative

Positive

Negative

Positive

Positive

Negative

Positive

- y : Gold standards / Gold labels / Ground truth
- $h(\vec{x})$: Predicted labels
 - Positives and Negatives

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8	75	127	160	95	Yes	No
9	66	66	135	90	Yes	Yes

True	Negative
False	Positive
True	Negative
True	Positive
False	Negative
False	Positive
True	Positive
False	Negative
True	Positive

- y : Gold standards / Gold labels / Ground truth
- $h(\vec{x})$: Predicted labels
 - Positives and Negatives
 - True (matching the ground truth) or False (not matching the ground truth)

The Output of a Classifier

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8	75	127	160	95	Yes	No
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True Negative

False Positive

Type-I Error

True Negative

True Positive

False Negative

Type-II Error

False Positive

Type-I Error

True Positive

False Negative

Type-II Error

True Positive

- y : Gold standards / Gold labels / Ground truth
- $h(\vec{x})$: Predicted labels
 - Positives and Negatives
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True Negative

False

Positive

Type-I Error

True Negative

True Positive

False

Negative

Type-II Error

False

Positive

Type-I Error

True Positive

False

Negative

Type-II Error

True Positive

Type-I errors

Type-II errors

gold standard labels

gold positive

gold negative

system
output
labels

system
positive

system
negative

true positive

false positive

false negative

true negative



LUMS



Evaluation of Classifiers

Performance Measures

Agha Ali Raza



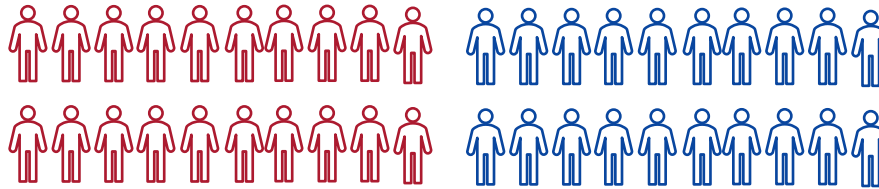
		Gold Labels	
		Gold Positive	Gold Negative
Predicted Labels	Predicted Positive	True Positives (tp)	False Positives (fp)
	Predicted Negative	False Negatives (fn)	True Negatives (tn)

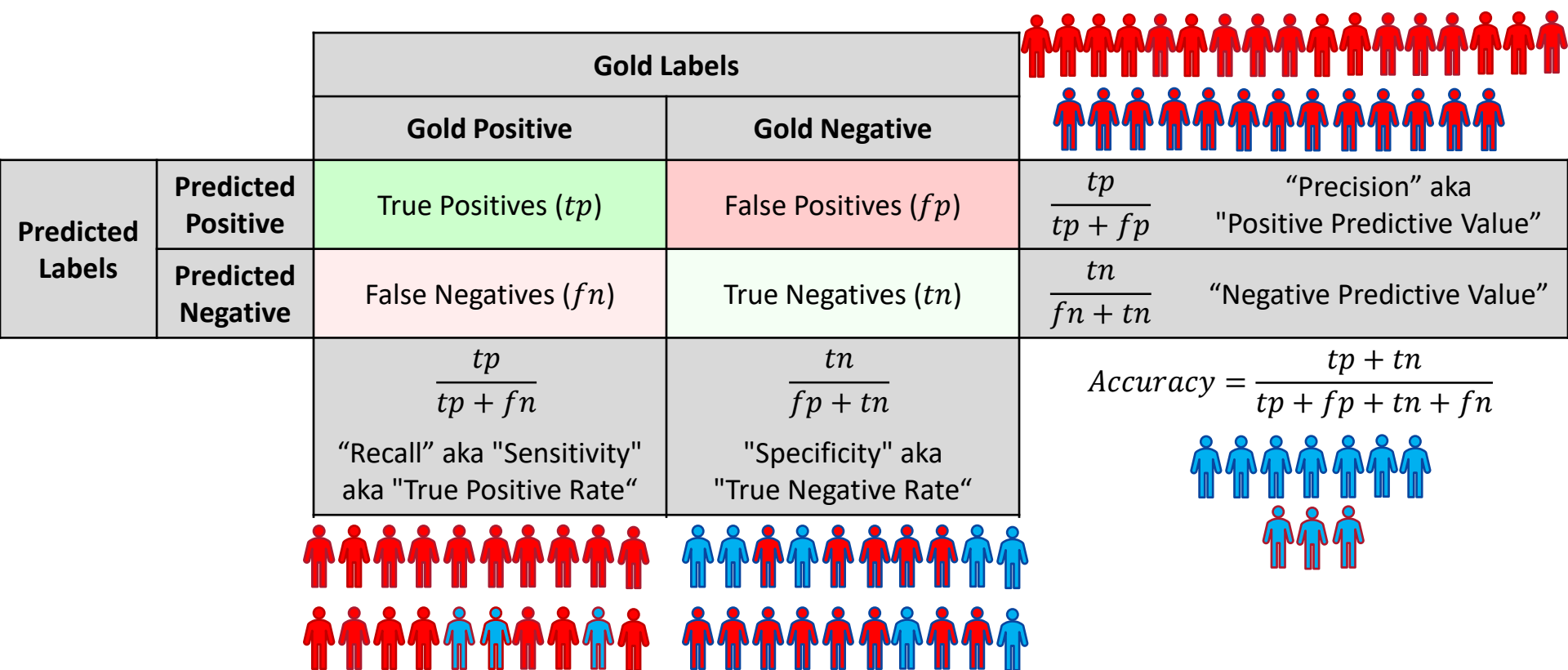
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		Gold Positive	Gold Negative
Predicted Labels	Predicted Positive	True Positives (<i>tp</i>)	False Positives (<i>fp</i>)
	Predicted Negative	False Negatives (<i>fn</i>)	True Negatives (<i>tn</i>)

$$Accuracy = \frac{tp + tn}{tp + fp + tn + fn}$$

- You are the CEO of a Pie Company and you need to know what people are saying about your pies on social media
- You build a system that detects tweets concerning your pies
 - the positive class is tweets about your pies
 - the negative class is all other tweets
- Imagine that we looked at a million tweets
 - 100 of them are discussing their love (or hatred) for our pie
 - the other 999,900, are tweets about other matters
- You created a classifier that stupidly classified every tweet as “not about pie”
 - 999,900 true negatives, 100 false negatives, 0 false positives, and 0 true positives, for an accuracy of 999,900/1,000,000 or 99.99%!
- **Accuracy is not a good metric when the goal is to discover a rare event, or at least not completely balanced in frequency**
- **A very common situation in the world!**

		Gold Labels			
		Gold Positive	Gold Negative		
Predicted Labels	Predicted Positive	True Positives (tp)	False Positives (fp)	$\frac{tp}{tp + fp}$	"Precision" aka "Positive Predictive Value"
	Predicted Negative	False Negatives (fn)	True Negatives (tn)	$\frac{tn}{fn + tn}$	"Negative Predictive Value"
		$\frac{tp}{tp + fn}$ "Recall" aka "Sensitivity" aka "True Positive Rate"	$\frac{tn}{fp + tn}$ "Specificity" aka "True Negative Rate"	$Accuracy = \frac{tp + tn}{tp + fp + tn + fn}$	





Performance Metrics

$$\text{Accuracy} = \frac{\text{My Correct Answers}}{\text{All Questions}} = \frac{tp + tn}{tp + tn + fp + fn}$$

- What fraction of time am I correct in my classification

$$\text{Precision} = \frac{\text{True Positives}}{\text{My Positives}} = \frac{tp}{tp + fp}$$

- How much should you trust me when I say that something tests positive
- What fraction of my positives are true positives

$$\text{Recall} = \text{Sensitivity} = \frac{\text{True Positives}}{\text{Real Positives}} = \frac{tp}{tp + fn}$$

- How much of the reality has been covered by my positive output?
- What fraction of the true positives is captured by my positives? E.g. How many sick people are correctly identified as having the condition?

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{Real Negatives}} = \frac{tn}{tn + fp}$$

- How much of the reality has been covered by my negative output?
- What fraction of the true negatives is captured by my negatives? E.g. How many identified healthy people do not have the condition?

A worked example

You are shown a set of 21 coins: 10 gold and 11 copper. Your task to accept all gold coins and reject all copper ones

You accept 7 coins as being gold (these are your positives)

- 5 of these are actually gold (these are your true positives, tp)
- 2 of these are copper (these are your false positives, fp)
- You falsely rejected 5 gold ones (false negatives, fn)
- You correctly rejected 9 copper ones (true negatives, tn)

	Actual Gold	Actual Copper
Predicted Gold	5	2
Predicted Copper	5	9

- Your **precision** is $\frac{tp}{\text{all of your positives}} = \frac{tp}{tp+fp} = \frac{5}{7}$
- Your **recall** is $\frac{tp}{\text{number of actual gold coins}} = \frac{tp}{tp+fn} = \frac{5}{10}$
- Your **specificity** is $\frac{tn}{\text{number of copper coins}} = \frac{tn}{fn+tn} = \frac{9}{11}$
- Your **accuracy** is $\frac{\text{correct answers}}{\text{all attempts}} = \frac{tp+tn}{tp+tn+fp+fn} = \frac{5+9}{5+9+2+5} = \frac{14}{21}$

Realistic extremes

You accept only one coin and that is gold

- Your precision is very high (1/1) but recall is very low (1/10)

	Ac Gld	Ac Cop
Pr Gld	1	0
Pr Cop	9	11

You return all 21 coins

- Your recall is very high (10/10) but precision is very low (10/21)

	Ac Gld	Ac Cop
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Only one out of the 21 coins is gold. And you reject everything.

- Your accuracy is very high ($20/21 = 0.95$) but precision/recall are 0

	Ac Gld	Ac Cop
Pr Gld	0	0
Pr Cop	1	20

So, what do we do now?

- A combined measure?

		Gold Labels			
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		$\frac{tp}{tp + fn}$ "Recall" aka "Sensitivity" aka "True Positive Rate"	$\frac{tn}{fp + tn}$ "Specificity" aka "True Negative Rate"	$Accuracy = \frac{tp + tn}{tp + fp + tn + fn}$	
		$\frac{fn}{tp + fn}$ 1 - Sensitivity = "False Negative Rate" aka "False Rejection Rate"	$\frac{fp}{fp + tn}$ 1 - Specificity = "False Positive Rate" aka "False Acceptance Rate"		

True Positive Rate = "Positive Likelihood Ratio"

False Positive Rate

False Negative Rate = "Negative Likelihood Ratio"

True Negative Rate

Probability = "Odds," often expressed as X:Y

1 - Probability

Evaluation of Classifiers

The F-Score

Agha Ali Raza



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True Positive Rate = "Positive Likelihood Ratio"

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So, what do we do now?

- **A combined measure?**

How to Combine Precision and Recall?

- The ratio: $\frac{Precision}{Recall}$ may work but does not have a way of differentially weighting Precision and Recall
- How about we take a mean of the two?
 - But what kind of a mean should we use?
 - Simple Arithmetic Mean is problematic: e.g.
 - $P = 0.0, R = 1.0, AM = 0.5$
 - $P = 0.1, R = 0.9, AM = 0.5$
 - $P = 0.5, R = 0.5, AM = 0.5$
- Requirements:
 - We need a weighted mean as we may care more about P or R
 - We need a conservative (deliberately lower) estimate of mean
 - If P and R are far apart, we need the mean to tend to the lower value
 - In order to do well, a classifier must do well on both P and R so that it cannot beat the system by being either too selective or too indiscriminate

Types of Means

Arithmetic Mean

$$AM = \frac{a_1 + a_2 + a_3 + \cdots + a_n}{n}$$

For 2 values: $AM = \frac{a_1 + a_2}{2}$

Geometric Mean

$$GM = \sqrt[n]{a_1 \cdot a_2 \cdot a_3 \cdots a_n}$$

For 2 values: $GM = \sqrt[2]{a_1 a_2}$

Harmonic Mean

$$HM = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \cdots + \frac{1}{a_n}}$$

For 2 values: $HM = \frac{2}{\frac{1}{a_1} + \frac{1}{a_2}} = \frac{2a_1 a_2}{a_1 + a_2}$

Harmonic Mean

Ref: <http://economistatlarge.com/finance/applied-finance/differences-arithmetic-geometric-harmonic-means>

- The harmonic mean of a set of numbers is the reciprocal of the arithmetic mean of reciprocals
- Best used in situations where extreme outliers exist in the population

x0	x1	x2	x3	x4	x5	x6	x7	x8	AM	GM	HM
1	2	3	4	5	6	7	8	9	5.00	4.15	3.18
2	4	8	16	32	64	128	256	512	113.56	32.00	9.02
5	5	5	5	5	5	5	5	5	5.00	5.00	5.00
5	5	5	5	5	5	5	5	10	5.56	5.40	5.29
5	5	5	5	5	5	5	5	100	15.56	6.97	5.59
5	5	5	5	5	5	5	5	1000	115.56	9.01	5.62
5	5	5	5	5	5	5	5	10000	1115.56	11.63	5.62
5	5	5	5	5	5	5	5	100000	11115.56	15.03	5.62
5	5	5	5	5	5	5	100000	100000	22226.11	45.16	6.43
5	5	5	5	5	100000	100000	100000	100000	44447.22	407.89	9.00
5	100000	100000	100000	100000	100000	100000	100000	100000	88889.44	33274.21	44.98
100000	100000	100000	100000	100000	100000	100000	100000	100000	100000.0	100000.0	100000.0

F-1-MEASURE

- The harmonic mean of P and R:
 - Is high when both P and R are high.
 - Is low when even one of P and R is low.
- A combined measure that assesses the P/R tradeoff is F measure (weighted harmonic mean of precision and recall):

$$F = \frac{2}{\frac{1}{P} + \frac{1}{R}} = \frac{2PR}{P + R}$$

F-MEASURE

$$F_{\beta} = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

The β parameter differentially weights the importance of recall and precision

- based on the needs of an application
- values of $\beta > 1$ favor recall, while
- values of $\beta < 1$ favor precision

When $\beta = 1$, precision and recall are equally balanced

This is the most frequently used metric, and is called $F_{\beta=1}$ or just $F1$:

$$F_1 = \frac{2PR}{P + R}$$

F- β -MEASURE

We can choose to favor precision or recall by using an interpolation weight α :

$$F = \frac{1}{\alpha \frac{1}{P} + (1-\alpha) \frac{1}{R}}$$
$$= \frac{(\beta^2 + 1)PR}{\beta^2 P + R} \quad \beta = \frac{1-\alpha}{\alpha}$$

$$F = \frac{1}{\left(\frac{1}{\beta^2 + 1} \frac{1}{P} + \frac{\beta^2}{\beta^2 + 1} \frac{1}{R} \right)}$$

$$F_\beta = \frac{1}{\frac{1}{\beta+1} \frac{1}{\text{precision}} + \frac{\beta}{\beta+1} \frac{1}{\text{recall}}}$$
$$= (1 + \beta) \frac{\text{precision} \cdot \text{recall}}{\beta \cdot \text{precision} + \text{recall}}$$

- Balanced F1 measure has $\beta = 1$ (that is, $\alpha = 1/2$) as shown above.
- To give more weight to the Precision, we pick a β value in the interval
 - $0 < \beta < 1$.
 - notice that it is getting multiplied with P in the denominator
- To give more weight to the Recall, we pick a β Value in the interval
 - $1 < \beta < +\infty$

F- β -MEASURE

We can choose to favor precision or recall by using an interpolation weight α :

$$F = \frac{1}{\alpha \frac{1}{P} + (1-\alpha) \frac{1}{R}}$$

$$= \frac{(\beta^2 + 1)PR}{\beta^2 P + R} \quad \beta = \frac{1-\alpha}{\alpha}$$

$$F = \frac{1}{\left(\frac{1}{\beta^2 + 1} \frac{1}{P} + \frac{\beta^2}{\beta^2 + 1} \frac{1}{R} \right)}$$

$$F_\beta = \frac{1}{\frac{1}{\beta+1} \frac{1}{\text{precision}} + \frac{\beta}{\beta+1} \frac{1}{\text{recall}}}$$

$$= (1 + \beta) \frac{\text{precision} \cdot \text{recall}}{\beta \cdot \text{precision} + \text{recall}}$$

- $\beta \rightarrow 0$ considers only precision

$$\frac{(\beta^2 + 1)PR}{\beta^2 P + R} = \frac{\beta^2 PR + PR}{\beta^2 P + R} = \frac{PR}{R} = P$$

- $\beta \rightarrow +\infty$ only recall

$$\frac{1}{\frac{1}{\beta+1} \frac{1}{P} + \frac{\beta}{\beta+1} \frac{1}{R}} = \frac{1}{0 \frac{1}{P} + 1 \frac{1}{R}} = R$$

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} \rightarrow 1 \text{ as } \frac{1}{2} \frac{2}{3} \frac{3}{4} \frac{4}{5} \frac{5}{6} \frac{6}{7} \frac{7}{8} \dots \text{ gets closer to 1}$$

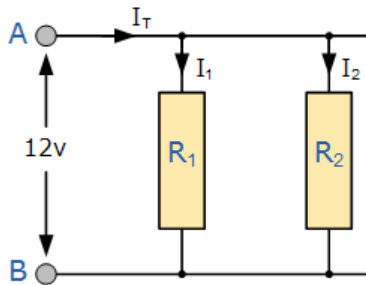
F-1, F-0.5, F-2

Precision	Recall	F-1	F-0.5	F-2
0	1	0	0	0
0.1	0.9	0.18	0.121622	0.346154
0.2	0.8	0.32	0.235294	0.5
0.3	0.7	0.42	0.33871	0.552632
0.4	0.6	0.48	0.428571	0.545455
0.5	0.5	0.5	0.5	0.5
0.6	0.4	0.48	0.545455	0.428571
0.7	0.3	0.42	0.552632	0.33871
0.8	0.2	0.32	0.5	0.235294
0.9	0.1	0.18	0.346154	0.121622
1	0	0	0	0
1	1	1	1	1
0.5	1	0.666667	0.555556	0.833333
1	0.5	0.666667	0.833333	0.555556
0.1	1	0.181818	0.121951	0.357143
1	0.1	0.181818	0.357143	0.121951

Intuition

Ref: <https://stats.stackexchange.com/questions/49226/how-to-interpret-f-measure-values>, http://www.electronics-tutorials.ws/resistor/res_4.html

The formula for F-measure (F1, with $\beta=1$) is the same as the formula giving the equivalent resistance composed of two resistances placed in parallel in physics (forgetting about the factor 2).



$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2}$$

This analogy would define F-measure as the equivalent resistance formed by *sensitivity* and *precision* placed in parallel.

Net resistance gets reduced as soon as any one among the two loose resistance

Evaluation of Classifiers

More than 2 Classes

Agha Ali Raza



Sources

- Jurafsky and Martin, SLP3,
<https://web.stanford.edu/~jurafsky/slp3/>

More than two classes

Lots of classification tasks in language processing have more than two classes:

- Sentiment analysis (positive, negative, neutral),
- Part-of-speech tagging (|POS tags|)
- Word sense disambiguation (|word senses|)
- Emotion detection (|emotions|)

More Than Two Classes: Sets of binary classifiers

Any-of or multi-label classification

- An instance can belong to 0, 1, or more than one classes.

For each class $c \in C$

- Build a binary classifier h_c to distinguish c from all other classes $c' \in C$ (c vs not c)

Given test instance d ,

- Evaluate it for membership in each class using each h_c
- Each h_c creates a score s_c , and returns 1 if s_c is greater than the defined threshold
- d belongs to any class for which h_c returns true

More Than Two Classes: Sets of binary classifiers

One-of or multinomial classification

- Classes are mutually exclusive: each instance in exactly one class

For each class $c \in C$

- Build a classifier h_c to distinguish c from all other classes $c' \in C$.

Given test instance d ,

- Evaluate it for membership in each class using each h_c
- d belongs to the **one** class with maximum score

Evaluation

3-way **one-of** email categorization decision (urgent, normal, spam)

		<i>gold labels</i>			
		urgent	normal	spam	
<i>system output</i>	urgent	8	10	1	precision_u = $\frac{8}{8+10+1}$
	normal	5	60	50	precision_n = $\frac{60}{5+60+50}$
	spam	3	30	200	precision_s = $\frac{200}{3+30+200}$
		recall_u = $\frac{8}{8+5+3}$	recall_n = $\frac{60}{10+60+30}$	recall_s = $\frac{200}{1+50+200}$	

Figure 6.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2), how many documents from c_1 were (in)correctly assigned to c_2

Micro- vs. Macro-Averaging

If we have more than one class, how do we combine multiple performance measures into one quantity?

1. **Macro-averaging:** Compute performance for each class, then average.
2. **Micro-averaging:** Collect decisions for all classes, compute contingency table, evaluate

Evaluation

		gold labels			
		urgent	normal	spam	
system output	urgent	8	10	1	$\text{precision}_u = \frac{8}{8+10+1}$
	normal	5	60	50	$\text{precision}_n = \frac{60}{5+60+50}$
	spam	3	30	200	$\text{precision}_s = \frac{200}{3+30+200}$
		$\text{recall}_u = \frac{8}{8+5+3}$	$\text{recall}_n = \frac{60}{10+60+30}$	$\text{recall}_s = \frac{200}{1+50+200}$	

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Macro Averaging

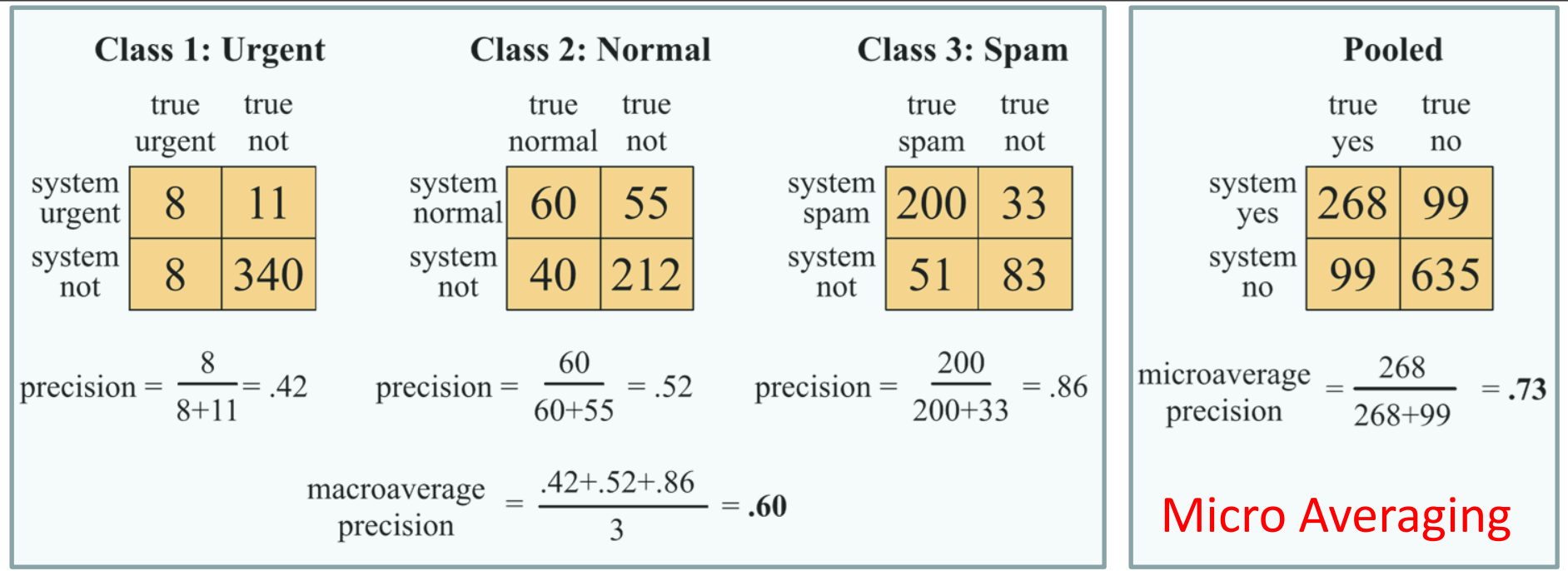


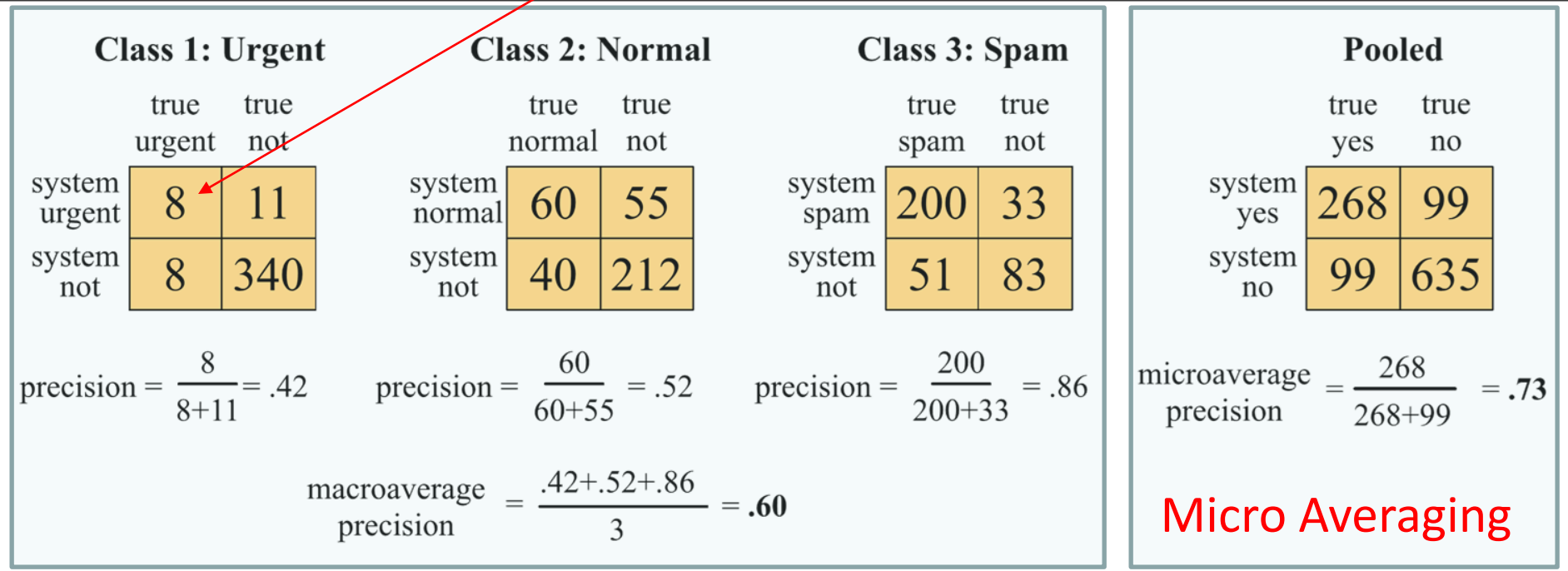
Figure 6.6 Separate contingency tables for the 3 classes from the previous figure, showing the pooled contingency table and the microaveraged and macroaveraged precision.

Evaluation

		gold labels			
		urgent	normal	spam	
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Macro Averaging



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	normal	5	60	50	$\text{precision}_n = \frac{60}{5+60+50}$
	spam	3	30	200	$\text{precision}_s = \frac{200}{3+30+200}$
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Figure 6.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2), how many documents from c_1 were (in)correctly assigned to c_2

Macro Averaging

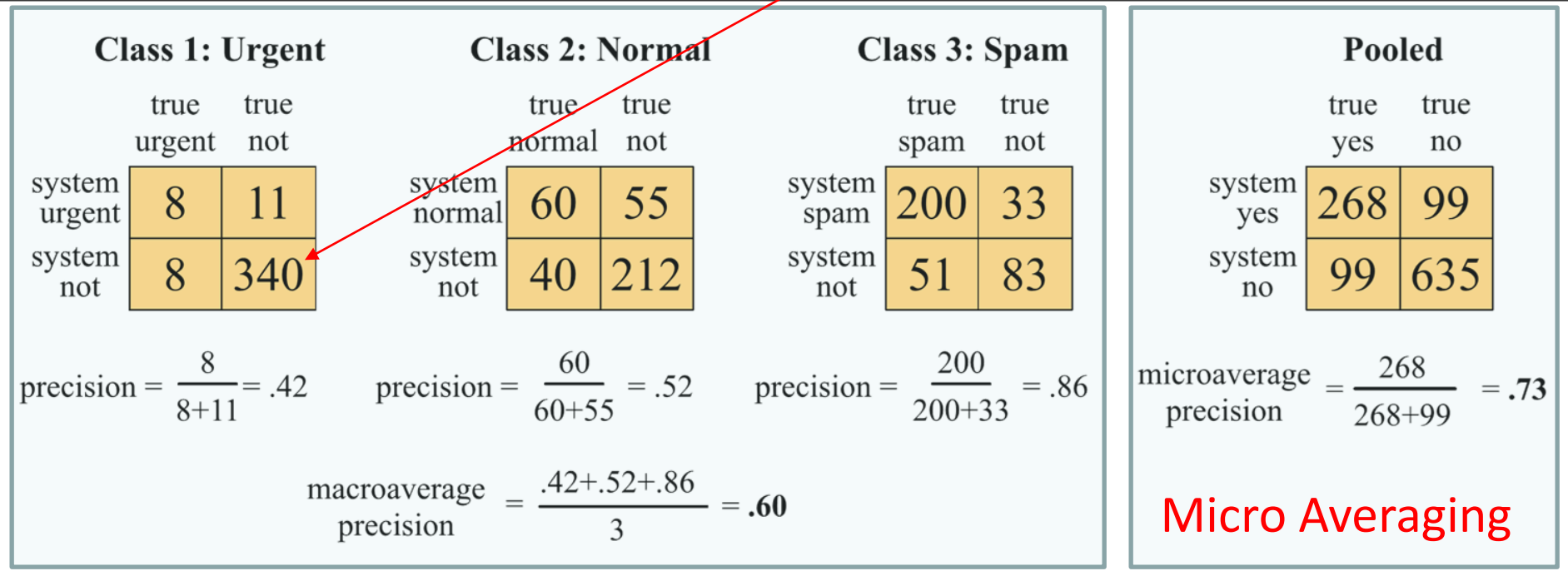


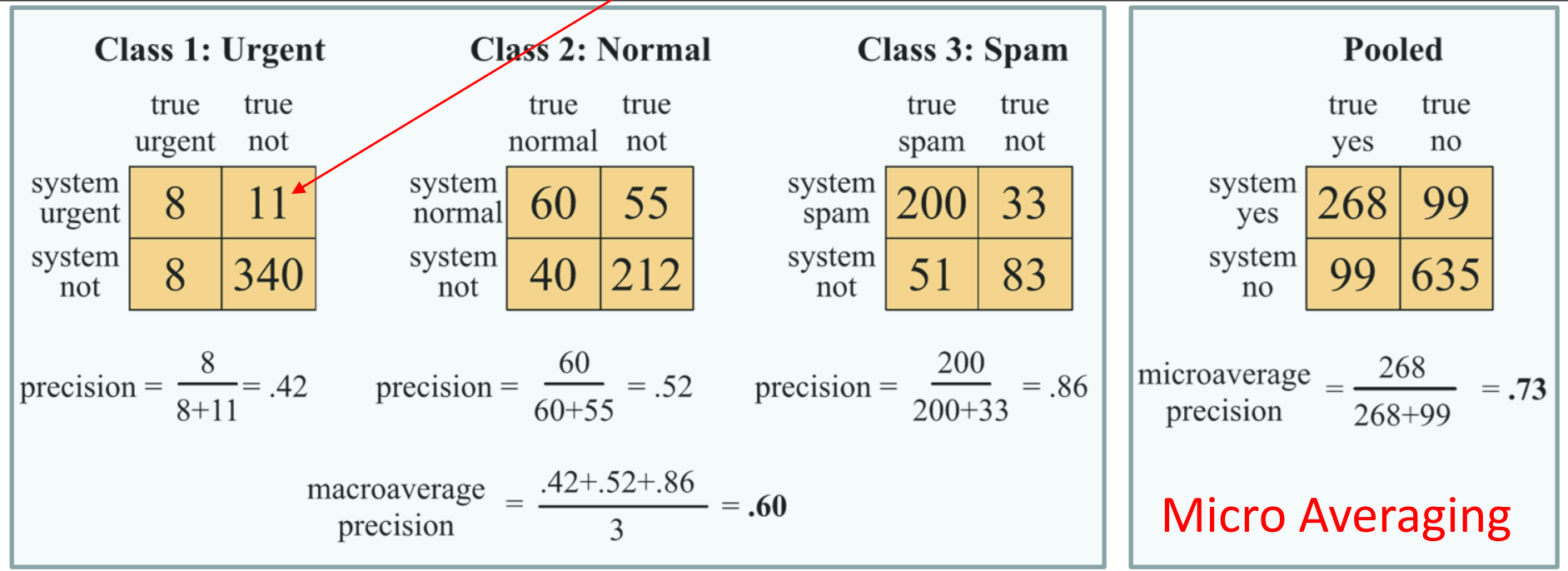
Figure 6.6 Separate contingency tables for the 3 classes from the previous figure, showing the pooled contingency table and the microaveraged and macroaveraged precision.

Evaluation

		gold labels			
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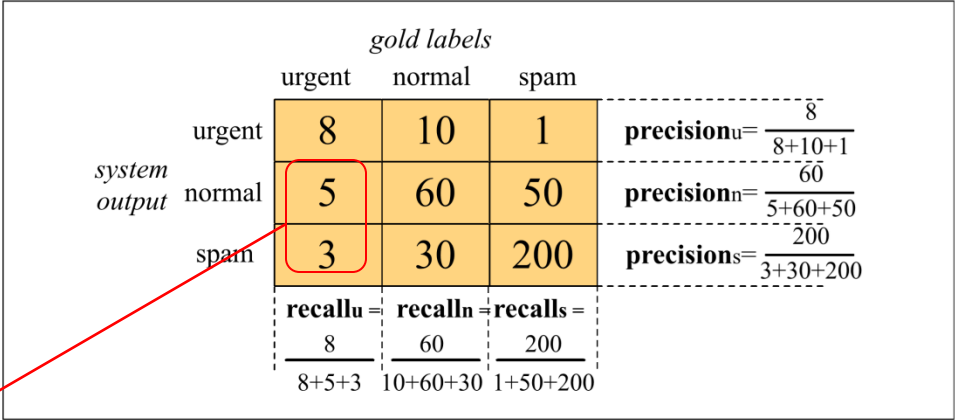
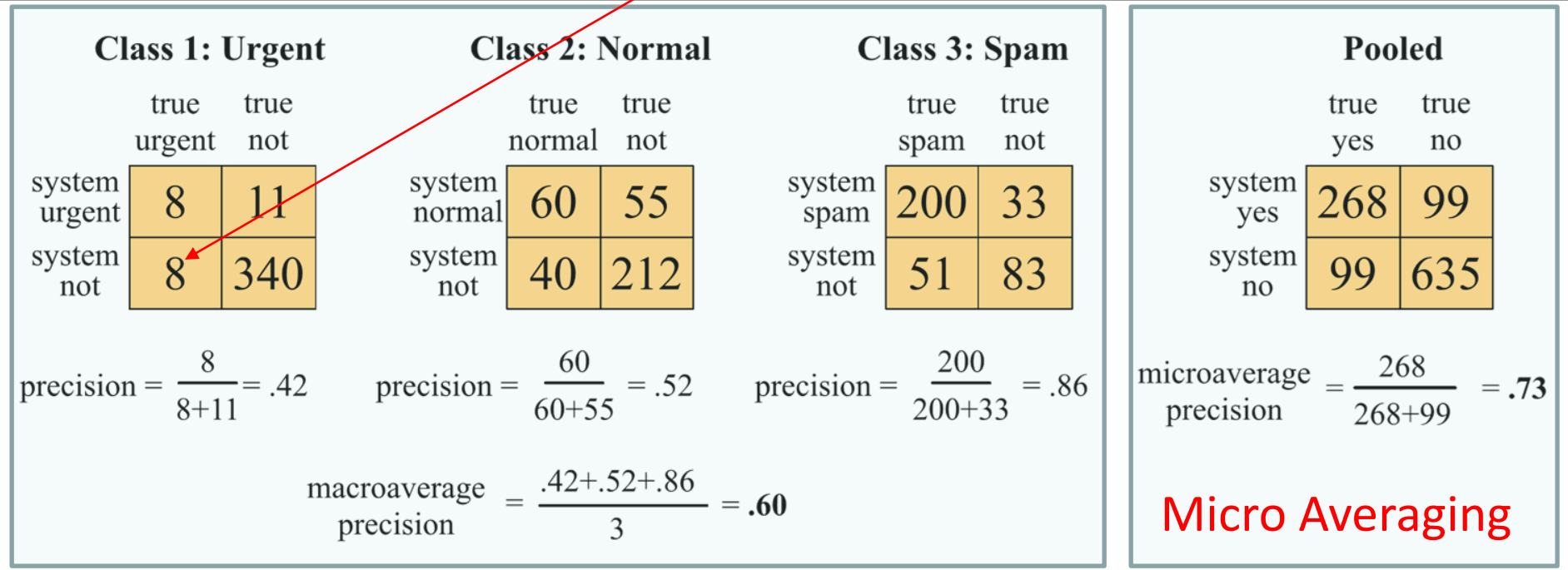


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Macro Averaging



Micro Averaging

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		gold labels			
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system output	urgent	8	10	1	$\text{precision}_u = \frac{8}{8+10+1}$
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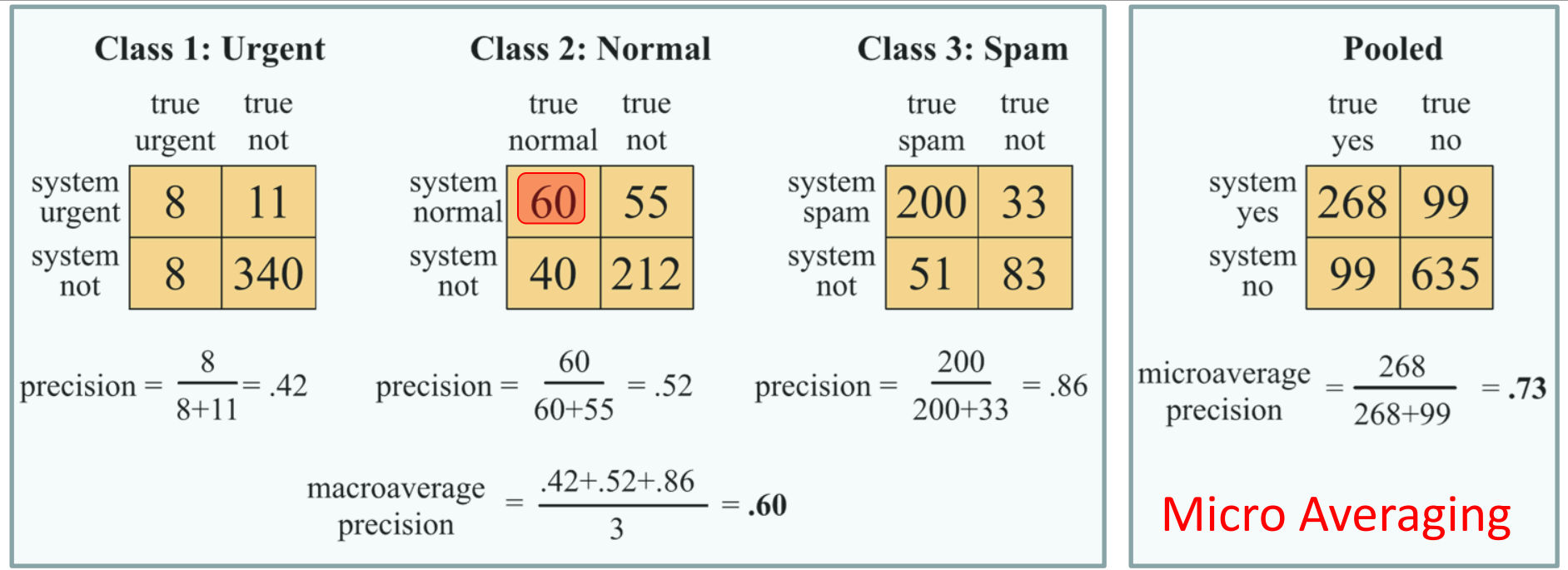


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Macro Averaging

Class 1: Urgent

	true urgent	true not
system urgent	8	11
system not	8	340

precision = $\frac{8}{8+11} = .42$

Class 2: Normal

	true normal	true not
system normal	60	55
system not	40	212

precision = $\frac{60}{60+55} = .52$

Class 3: Spam

	true spam	true not
system spam	200	33
system not	51	83

precision = $\frac{200}{200+33} = .86$

Pooled

	true yes	true no
system yes	268	99
system no	99	635

microaverage precision = $\frac{268}{268+99} = .73$

Micro Averaging

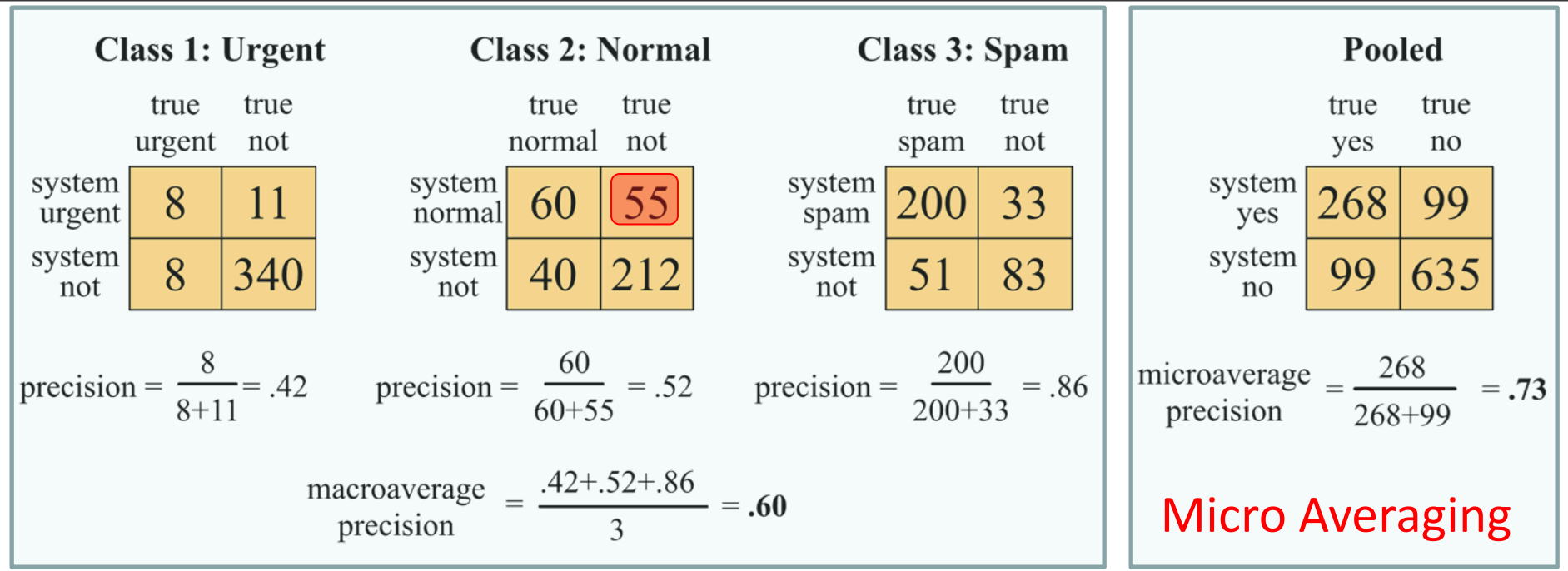
macroaverage precision = $\frac{.42+.52+.86}{3} = .60$

Evaluation

		gold labels			
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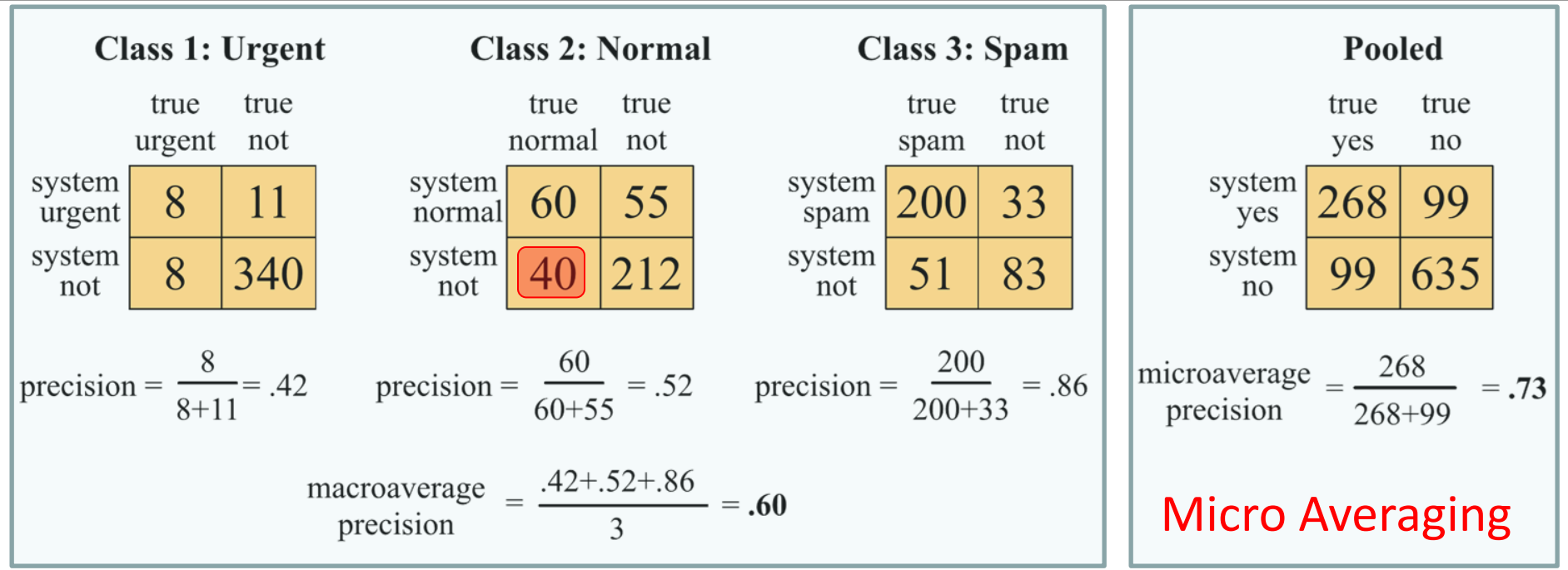


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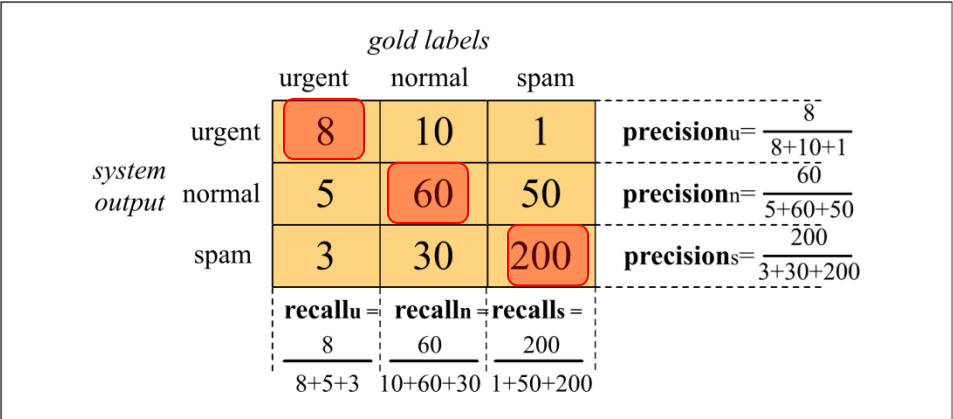
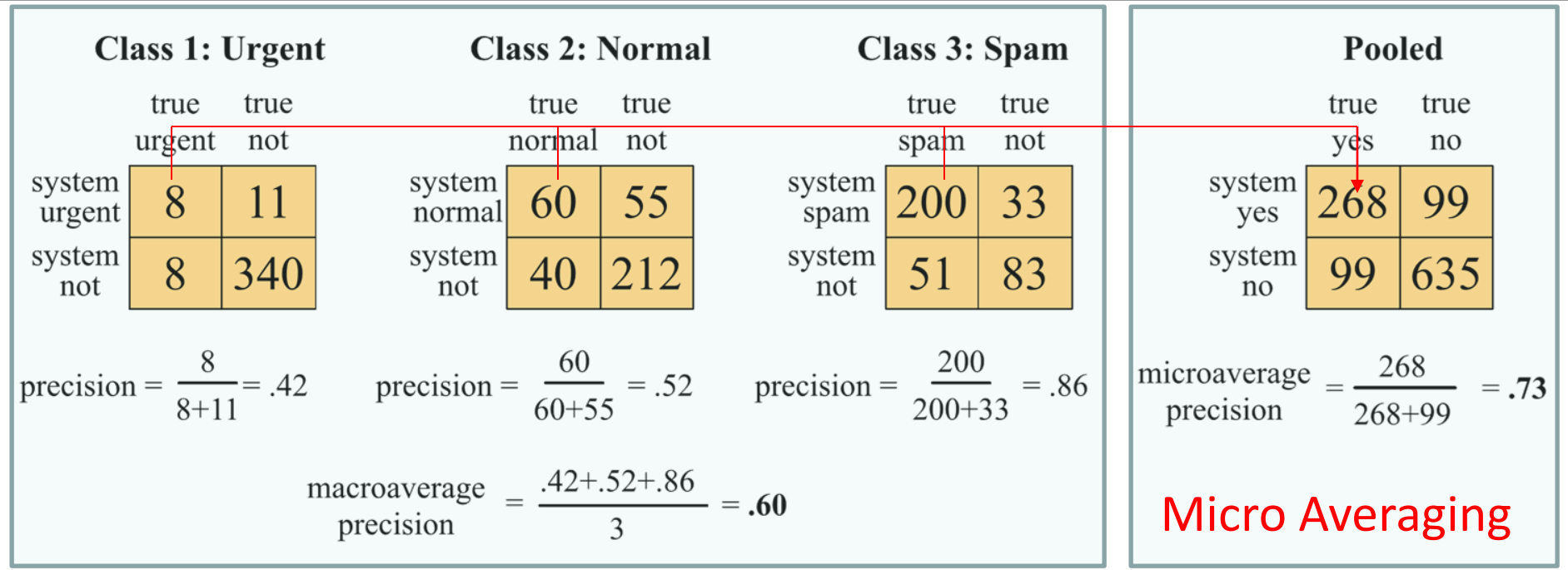


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Macro Averaging



Micro Averaging

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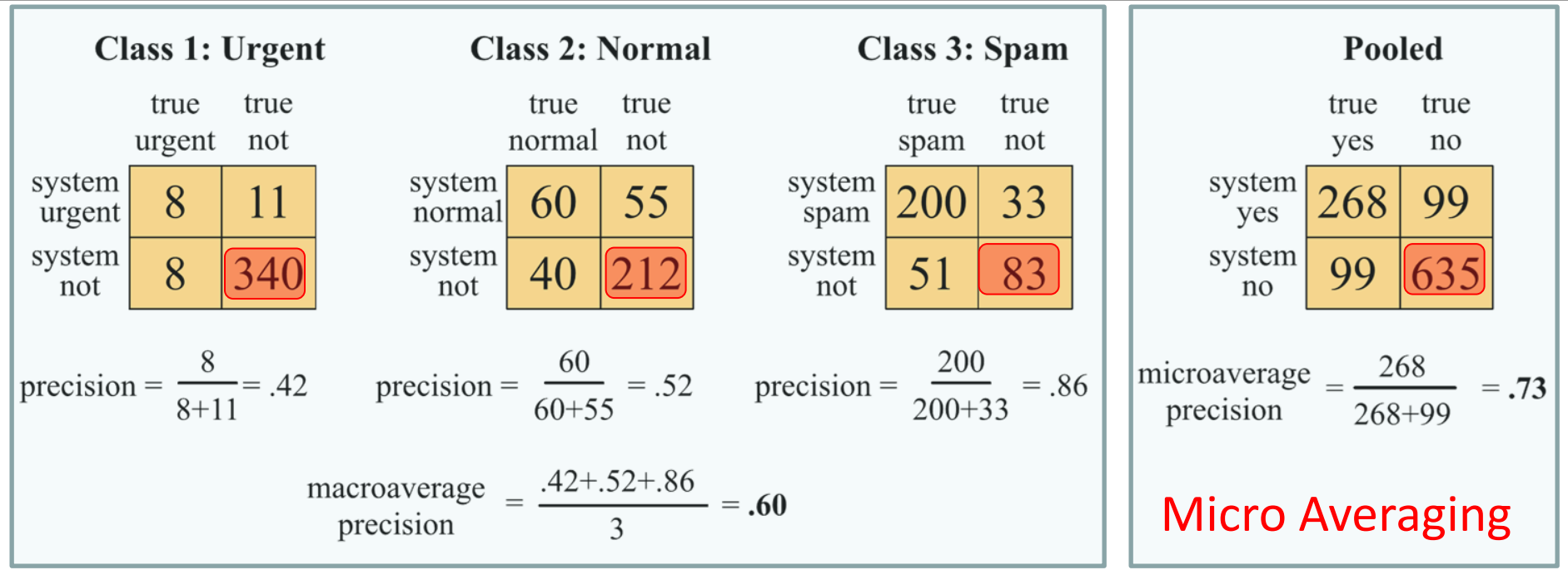


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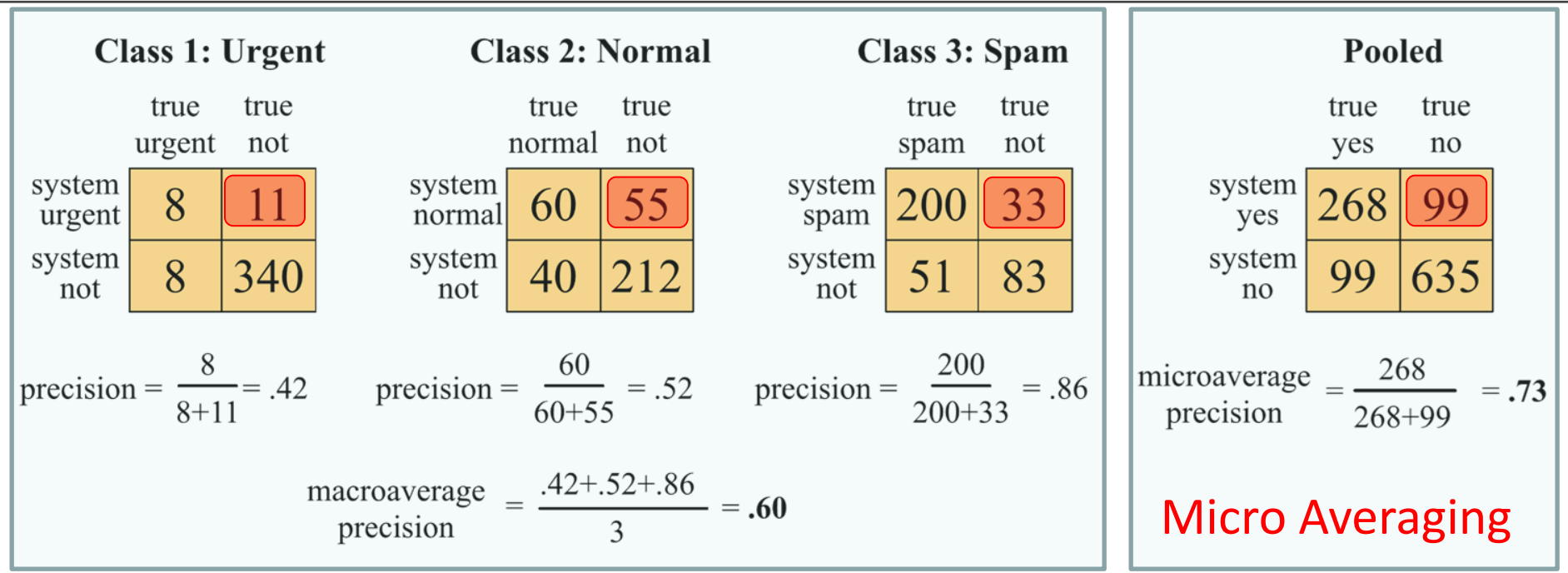


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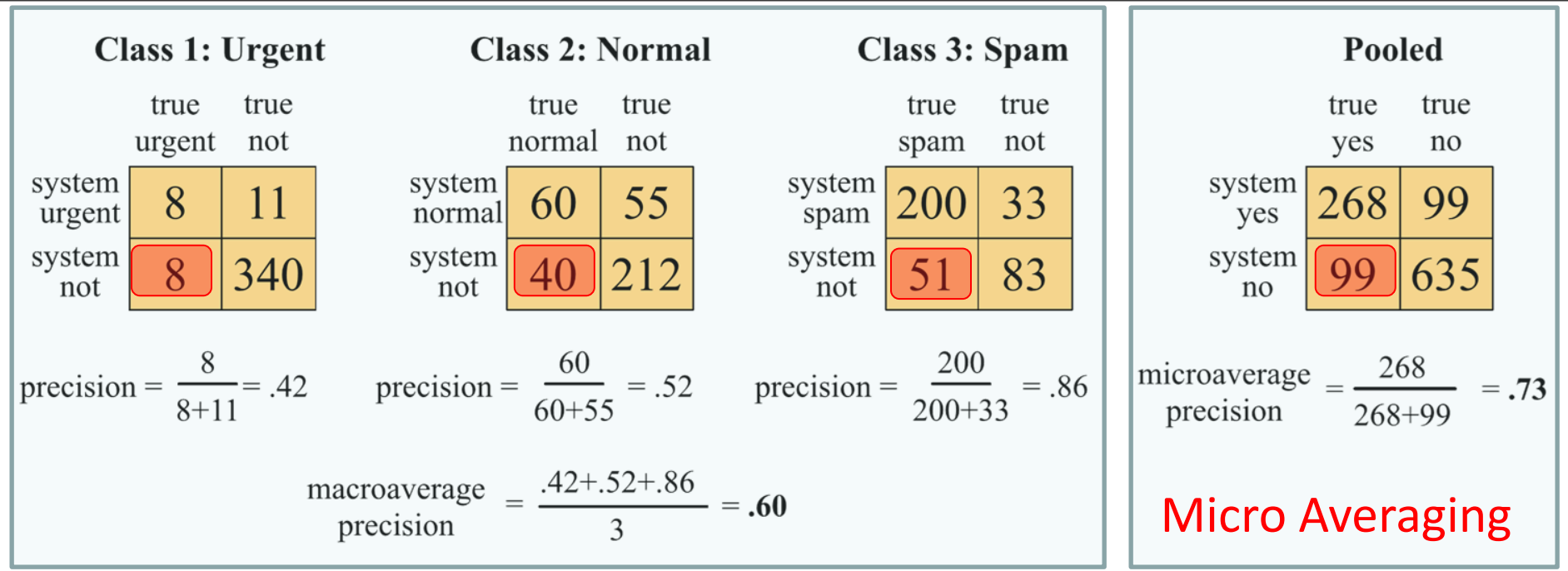


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Evaluation

A micro-average is dominated by the more frequent class (in this case spam)

- as the counts are pooled

The macro-average better reflects the statistics of the smaller classes,

- is more appropriate when performance on all the classes is equally important.

Evaluation of Classifiers

The Gold Labels

Agha Ali Raza



The Output of a Classifier

#	Height (inches)	Weight (kgs)	B.P. Sys	B.P. Dia	Heart disease	
	$\vec{x}$				y	$h(\vec{x})$
1	62	70	120	80	No	No
2	72	90	110	70	No	Yes
3	74	80	130	70	No	No
4	65	120	150	90	Yes	Yes
5	67	100	140	85	Yes	No
6	64	110	130	90	No	Yes
7	69	150	170	100	Yes	Yes
8	75	127	160	95	Yes	No
9	66	66	135	90	Yes	Yes

- y : Gold standards / Gold labels / Ground truth
- $h(\vec{x})$: Predicted labels

(How) can we trust the Gold Labels?

- Objective data gathered from the real world:
 - Measurement of heights, weights, weather phenomena
 - Aberrations, sensor malfunctions and errors
- Subjective data, annotated by humans (experts or community)
 - Tagging emotions, SPAM/HAM, sentiment, misinformation
 - Errors of judgement, biases, human error
- Use multiple sources for each label
 - Multiple sensors measuring the same phenomenon
 - Multiple humans annotating the same data
 - Inter-annotator agreements
 - Resource intense
 - Partial overlap of data
- Is annotator agreement the answer to all our worries?
 - Distribution of annotators
 - Chance agreements

Intuition of Agreement (and Randomness)

- **Annotator1 (A1)** and **Annotator2 (A2)** are deciding if posts should be deleted from a social media platform.
- Most of the posts should not be deleted.
- A1 is not an expert and **just randomly selects** 1% of posts.
- A2 is an expert and very carefully selects 1% of posts.
- A1 and A2 still agree on at least 98% of posts (that need not be deleted). Yaayyy!! ??
- **So, what is the problem?**
- The problem is that A1 was just picking posts at random. Because most posts are not deleted, A1 and A2 agree most of the time.
- But that does not mean that A1 and A2 are using similar ratings.
- Cohen's Kappa corrects for this – It would be approximately zero in this case.

<https://www.quora.com/What-is-an-intuitive-explanation-of-Cohens-kappa-statistic>

Gold Labels, Annotators and Agreement

- Multiple annotators

Utterances	Ann1	Ann2	Raw Agreement	Ann Rand	Agreement (A1, Rand)	Agreement (A2, Rand)
S1	+	+	1	-	0	0
S2	-	-	1	+	0	0
S3	+	-	0	+	1	0
S4	-	+	0	-	1	0
...	...	...	...	...	...	...
			Sum of Agreements		Sum of chance agreements	Sum of chance agreements

- Raw (observed) agreements
- Chance agreements
- Cohen's Kappa* ($\kappa = \frac{P_o - P_e}{1 - P_e} = 1 - \frac{1 - P_o}{1 - P_e}$) and *Krippendorff's alpha* ($\alpha = 1 - \frac{D_o}{D_e}$, where D is disagreement)
- Agreement over a random subset with an expert annotator

<https://towardsdatascience.com/cohens-kappa-9786ceceab58#:~:text=Cohen's%20kappa%20measures%20the%20agreement,raters%20may%20agree%20by%20chance.>
[https://repository.upenn.edu/cgi/viewcontent.cgi?article=1043&context=asc_papers#:~:text=Krippendorff's%20alpha%20\(%CE%B1\)%20is%20a,assign%20computable%20values%20to%20them.](https://repository.upenn.edu/cgi/viewcontent.cgi?article=1043&context=asc_papers#:~:text=Krippendorff's%20alpha%20(%CE%B1)%20is%20a,assign%20computable%20values%20to%20them.)

Cohen's Kappa

$$\kappa = \frac{P_o - P_e}{1 - P_e} = 1 - \frac{1 - P_o}{1 - P_e}$$

- P_o is the probability of actual agreement
- P_e is the expected probability of “agreement by chance” - the probability of the agreement if the rating is random
- The numerator is the **difference between the observed agreement and chance agreement**
- The denominator is the **difference between the maximum possible observed agreement (i.e., 1: 100% agreement) and the chance agreement**
- As the denominator is just a normalizer, observe that κ is high when the difference between the observed and chance agreement is the highest
 - For a good model, the observed difference and the maximum difference are close to each other, and Cohen's kappa is close to 1.
 - For a random model, the overall accuracy is all due to random chance, the numerator is 0, and Cohen's kappa is 0.
 - Cohen's kappa could also theoretically be negative. Then, the overall accuracy of the model would be even lower than what could have been obtained by a random guess.

Kappa value interpretation Landis & Koch (1977):

<0 No agreement

0 – .20 Slight

.21 – .40 Fair

.41 – .60 Moderate

.61 – .80 Substantial

.81–1.0 Perfect

<https://thenewstack.io/cohens-kappa-what-it-is-when-to-use-it-and-how-to-avoid-its-pitfalls/>

Cohen's Kappa

Suppose that you were analyzing data related to a group of 50 people applying for a grant. Each grant proposal was read by two readers and each reader either said "Yes" or "No" to the proposal. Suppose the disagreement count data were as follows, where A and B are readers.

- The observed proportionate agreement is:

$$p_o = \frac{a + d}{a + b + c + d} = \frac{20 + 15}{50} = 0.7$$

- To calculate p_e (the probability of random agreement) we note that:
 - Reader A said "Yes" to 25 applicants and "No" to 25 applicants. Thus reader A said "Yes" 50% of the time.
 - Reader B said "Yes" to 30 applicants and "No" to 20 applicants. Thus reader B said "Yes" 60% of the time.
- So the expected probability that both would say yes at random is:

$$p_{\text{Yes}} = \frac{a + b}{a + b + c + d} \cdot \frac{a + c}{a + b + c + d} = 0.5 \times 0.6 = 0.3$$

- Similarly,

$$p_{\text{No}} = \frac{c + d}{a + b + c + d} \cdot \frac{b + d}{a + b + c + d} = 0.5 \times 0.4 = 0.2$$

- Overall random agreement probability is the probability that they agreed on either Yes or No, i.e.:

$$p_e = p_{\text{Yes}} + p_{\text{No}} = 0.3 + 0.2 = 0.5$$

- So now applying our formula for Cohen's Kappa we get:

$$\kappa = \frac{p_o - p_e}{1 - p_e} = \frac{0.7 - 0.5}{1 - 0.5} = 0.4$$

		B	
		Yes	No
A	Yes	a	b
	No	c	d

		B	
		Yes	No
A	Yes	20	5
	No	10	15

Kappa value interpretation Landis & Koch (1977):

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From: https://en.wikipedia.org/wiki/Cohen%27s_kappa

Evaluation of Classifiers

Which Mean to Use?

Agha Ali Raza



		Gold Labels			
		Gold Positive	Gold Negative		
Predicted Labels	Predicted Positive	True Positives (<i>tp</i>)	False Positives (<i>fp</i>)	$\frac{tp}{tp + fp}$	"Precision" aka "Positive Predictive Value"
	Predicted Negative	False Negatives (<i>fn</i>)	True Negatives (<i>tn</i>)	$\frac{tn}{fn + tn}$	"Negative Predictive Value"
		$\frac{tp}{tp + fn}$ "Recall" aka "Sensitivity" aka "True Positive Rate"	$\frac{tn}{fp + tn}$ "Specificity" aka "True Negative Rate"	$Accuracy = \frac{tp + tn}{tp + fp + tn + fn}$	
		$\frac{fn}{tp + fn}$ 1 - Sensitivity = "False Negative Rate" aka "False Rejection Rate"	$\frac{fp}{fp + tn}$ 1 - Specificity = "False Positive Rate" aka "False Acceptance Rate"		

True Positive Rate = "Positive Likelihood Ratio"

False Positive Rate

False Negative Rate = "Negative Likelihood Ratio"

True Negative Rate

Probability = "Odds," often expressed as X:Y

1 - Probability

A worked example

You are shown a set of 21 coins: 10 gold and 11 copper. Your task to accept all gold coins and reject all copper ones

You accept 7 coins as being gold (these are your positives)

- 5 of these are actually gold (these are your true positives, tp)
- 2 of these are copper (these are your false positives, fp)
- You falsely rejected 5 gold ones (false negatives, fn)
- You correctly rejected 9 copper ones (true negatives, tn)

	Actual Gold	Actual Copper
Predicted Gold	5	2
Predicted Copper	5	9

- Your **precision** is $\frac{tp}{\text{all of your positives}} = \frac{tp}{tp+fp} = \frac{5}{7}$
- Your **recall** is $\frac{tp}{\text{number of actual gold coins}} = \frac{tp}{tp+fn} = \frac{5}{10}$
- Your **specificity** is $\frac{tn}{\text{number of copper coins}} = \frac{tn}{fn+tn} = \frac{9}{11}$
- Your **accuracy** is $\frac{\text{correct answers}}{\text{all attempts}} = \frac{tp+tn}{tp+tn+fp+fn} = \frac{5+9}{5+9+2+5} = \frac{14}{21}$

Realistic extremes

You accept only one coin and that is gold

- Your precision is very high (1/1) but recall is very low (1/10)

	Ac Gld	Ac Cop
Pr Gld	1	0
Pr Cop	9	11

You return all 21 coins

- Your recall is very high (10/10) but precision is very low (10/21)

	Ac Gld	Ac Cop
Pr Gld	10	11
Pr Cop	0	0

Only one out of the 21 coins is gold. And you reject everything.

- Your accuracy is very high ($20/21 = 0.95$) but precision/recall are 0

	Ac Gld	Ac Cop
Pr Gld	0	0
Pr Cop	1	20

So, what do we do now?

- A combined measure?

A combined measure of Precision and Recall

- It is useful to have a single number to describe performance. Should be high when both P and R are high.
- The mean of precision and recall?
- But what kind of a mean should we use?
- Simple Arithmetic Mean is problematic: e.g.
 - $P = 0.0, R = 1.0, AM = 0.5$
 - $P = 0.1, R = 0.9, AM = 0.5$
- **Requirements:**
 - We need a weighted mean as we may care more about P or R
 - We need a conservative (deliberately lower) estimate of mean
 - If P and R are far apart we need the mean to tend to the lower value
 - In order to do well, a classifier must do well on both P and R so that it cannot beat the system by being either too selective or too indiscriminate

What is mean (or average)?

- The **central tendency** is a single number that represents the most common value for a list of numbers.
 - It is the value that has the highest probability from the probability distribution that describes all possible values that a random variable may have.
- Many ways to calculate it
 - **Mode:** the most common value in the data distribution
 - **Median:** the middle value if all values in the data sample were ordered
 - **Mean:** the average value – Three common types
- The *mean* is different from the *median* and the *mode* in that it is a measure of the central tendency that is calculated from the data.

What is mean (or average)?

- Different ways to calculate the mean based on the type of data.
- Three common types:
 - Arithmetic mean
 - Geometric mean
 - Harmonic mean
aka, the Pythagorean means

Types of Means

Arithmetic Mean

$$AM = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$$

For 2 values: $AM = \frac{a_1 + a_2}{2}$

Geometric Mean

$$GM = \sqrt[n]{a_1 \cdot a_2 \cdot a_3 \dots a_n}$$

For 2 values: $GM = \sqrt[2]{a_1 a_2}$

Harmonic Mean

$$HM = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n}}$$

For 2 values: $HM = \frac{2}{\frac{1}{a_1} + \frac{1}{a_2}} = \frac{2a_1 a_2}{a_1 + a_2}$

Arithmetic Mean

Ref: <http://economistatlarge.com/finance/applied-finance/differences-arithmetic-geometric-harmonic-means>

- The most common average, simplest to compute
- The AM of 2, 3 and 4 is 3
- Suitable when:
 - the data is not skewed (no extreme outliers)
 - the individual data points are not dependent on each other
 - the numbers are relatively evenly distributed, e.g.
 - follow a normal distribution
 - when you are rolling a fair die, the expected value is the mean of all the numbers on it
- Easily distorted if the sample of observations contains outliers or for data that has a non-Gaussian distribution (e.g. multiple peaks – a multi-modal probability distribution).
- The AM is more meaningful when a variable has a Gaussian or Gaussian-like data distribution.

Geometric Mean

Ref: <https://towardsdatascience.com/on-average-youre-using-the-wrong-average-geometric-harmonic-means-in-data-analysis-2a703e21ea0>

- The geometric mean multiplies rather than sums values, then takes the n th root rather than dividing by n
- It essentially says: if every number in our dataset was the same number, what would that number have to be in order to have the same *multiplicative product* as our actual dataset?
- This makes it well-suited for describing multiplicative relationships, such as rates and ratios, even if those ratios are on different scales (i.e. do not have the same denominator)
- The geometric mean works well when the data is in an multiplicative relationship or in cases where the data is compounded

Geometric Mean

Ref: <https://towardsdatascience.com/on-average-youre-using-the-wrong-average-geometric-harmonic-means-in-data-analysis-2a703e21ea0>

- Geometric mean should be used when the data points are inter-related
- It's appropriate for numbers that are distributed along a logarithmic scale - that is, when you're as likely to find a number twice the size as a number half the size.
- The geometric mean does not accept negative or zero values, e.g. all values must be positive.
- A fancy feature of the geometric mean is that you can often average across numbers on *completely different scales*.

Geometric Mean

- Compare ratings for two coffeeshops using two different sources.
- The problem is that *source 1* uses a *5-star* scale and *source 2* uses a *100-point* scale:

Coffeeshop A

- Source 1 rating: 4.5
- Source 2 rating: 68

Coffeeshop B

- Source 1 rating: 3
- Source 2 rating: 75

If we naively take the arithmetic mean of raw ratings for each coffeeshop:

$$\text{Coffeeshop A} = (4.5 + 68) \div 2 = 36.25$$

$$\text{Coffeeshop B} = (3 + 75) \div 2 = 39$$

We'd conclude that **Coffeeshop B** was the winner.

Geometric Mean

- **The right way to do this is:**
- We need to normalize our values onto the same scale before averaging them with the arithmetic mean, to get an accurate result.
- So we multiply the source 1 ratings by 20 to bring them from a 5-star scale to the 100-point scale of source 2:

Coffeeshop A

- $4.5 * 20 = 90$
- $(90 + 68) \div 2 = 79$

Coffeeshop B

- $3 * 20 = 60$
- $(60 + 75) \div 2 = 67.5$

So we find that **Coffeeshop A** is the true winner, contrary to the naive application of arithmetic mean above.

Geometric Mean

- **For this particular problem**, the geometric mean, allows us to reach the same conclusion:
Coffeeshop A = square root of $(4.5 * 68) = 17.5$
Coffeeshop B = square root of $(3 * 75) = 15$
- The arithmetic mean is dominated by numbers on the larger scale, which makes us think Coffeeshop B is the higher rated shop. This is because the arithmetic mean expects an additive relationship between numbers and does not account for scales and proportions. Hence the need to bring numbers onto the same scale before applying the arithmetic mean.
- The geometric mean, on the other hand, can handle varying proportions with ease, due to it's multiplicative nature. This is a tremendously useful property, but **we no longer have any interpretable scale at all**. The geometric mean is effectively unitless in such situations.

Harmonic Mean

Ref: <http://economistatlarge.com/finance/applied-finance/differences-arithmetic-geometric-harmonic-means>

- The harmonic mean of a set of numbers is the reciprocal of the arithmetic mean of reciprocals
- Best used in situations where extreme outliers exist in the population

x0	x1	x2	x3	x4	x5	x6	x7	x8	AM	GM	HM
1	2	3	4	5	6	7	8	9	5.00	4.15	3.18
2	4	8	16	32	64	128	256	512	113.56	32.00	9.02
5	5	5	5	5	5	5	5	5	5.00	5.00	5.00
5	5	5	5	5	5	5	5	10	5.56	5.40	5.29
5	5	5	5	5	5	5	5	100	15.56	6.97	5.59
5	5	5	5	5	5	5	5	1000	115.56	9.01	5.62
5	5	5	5	5	5	5	5	10000	1115.56	11.63	5.62
5	5	5	5	5	5	5	5	100000	11115.56	15.03	5.62
5	5	5	5	5	5	5	100000	100000	22226.11	45.16	6.43
5	5	5	5	5	100000	100000	100000	100000	44447.22	407.89	9.00
5	100000	100000	100000	100000	100000	100000	100000	100000	88889.44	33274.21	44.98
100000	100000	100000	100000	100000	100000	100000	100000	100000	100000.0	100000.0	100000.0

A combined measure for Prec Rec

Use harmonic mean instead of arithmetic mean as we are taking the average of ratios

Harmonic mean *punishes* extreme values more strictly:

- Consider a *trivial* classifier (always returns class A)
- A very large number of data elements of class B, and a single element of class A:
 - Precision: 0.0
 - Recall: 1.0
 - Arithmetic mean = 0.5 (50% correct), despite being the *worst* possible outcome!
 - The harmonic mean is nearly 0.
 - To have a high F-score, you need *both* a high precision and high recall

F-1-MEASURE

A combined measure that assesses the P/R tradeoff is F measure (weighted harmonic mean):

$$F = \frac{2}{\frac{1}{P} + \frac{1}{R}} = \frac{2PR}{P + R}$$

F- β -MEASURE

We can choose to favor precision or recall by using an interpolation weight α :

$$F = \frac{1}{\alpha \frac{1}{P} + (1-\alpha) \frac{1}{R}}$$
$$= \frac{(\beta^2 + 1)PR}{\beta^2 P + R} \qquad \beta = \frac{1-\alpha}{\alpha}$$

- Balanced F1 measure has $\beta = 1$ (that is, $\alpha = 1/2$) as shown above.
- To give more weight to the Precision, we pick a β value in the interval
 - $0 < \beta < 1$.
 - Notice that it is getting multiplied with P in the denominator
- To give more weight to the Recall, we pick a β Value in the interval
 - $1 < \beta < +\infty$
- $\beta \rightarrow 0$ considers only precision, $\beta \rightarrow +\infty$ only recall

F-MEASURE

$$F_{\beta} = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

The β parameter differentially weights the importance of recall and precision

- based on the needs of an application
- values of $\beta > 1$ favor recall, while
- values of $\beta < 1$ favor precision

When $\beta = 1$, precision and recall are equally balanced

This is the most frequently used metric, and is called

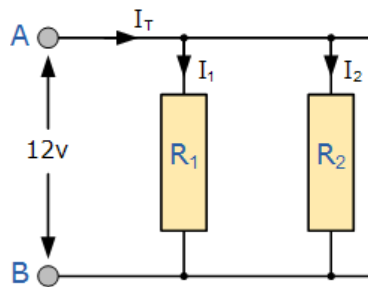
$F_{\beta=1}$ or just $F1$:

$$F_1 = \frac{2PR}{P + R}$$

Intuition

Ref: <https://stats.stackexchange.com/questions/49226/how-to-interpret-f-measure-values>, http://www.electronics-tutorials.ws/resistor/res_4.html

The formula for F-measure (F1, with $\beta=1$) is the same as the formula giving the equivalent resistance composed of two resistances placed in parallel in physics (forgetting about the factor 2).



$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2}$$

This analogy would define F-measure as the equivalent resistance formed by *sensitivity* and *precision* placed in parallel.

Net resistance gets reduced as soon as any one among the two loose resistance

Evaluation of Classifiers

ROC, PR Curves, AUC, EER

Agha Ali Raza



		Gold Labels			
		Gold Positive	Gold Negative		
Predicted Labels	Predicted Positive	True Positives (tp)	False Positives (fp)	$\frac{tp}{tp + fp}$	"Precision" aka "Positive Predictive Value"
	Predicted Negative	False Negatives (fn)	True Negatives (tn)	$\frac{tn}{fn + tn}$	"Negative Predictive Value"
		$\frac{tp}{tp + fn}$ "Recall" aka "Sensitivity" aka "True Positive Rate" "True Acceptance Rate"	$\frac{tn}{fp + tn}$ "Specificity" aka "True Negative Rate" "True Rejection Rate"	$Accuracy = \frac{tp + tn}{tp + fp + tn + fn}$	
		$\frac{fn}{tp + fn}$ 1 - Sensitivity = "False Negative Rate" aka "False Rejection Rate"	$\frac{fp}{fp + tn}$ 1 - Specificity = "False Positive Rate" aka "False Acceptance Rate"		

- Sensitivity, specificity, FNR and FPR are not influenced by real-world data imbalances
 - These imbalances impact the denominators
 - E.g. a rare disease, or a rare phenomena (like a fraud email)

	Actual +	Actual -
Test +	5	5,000
Test -	5	5,000

$Sensitivity = \frac{5}{10} = 0.5$,
 $Precision = \frac{5}{5+5000} = 0.0009$

- Precision and Negative Predictive Value are impacted by these imbalances and are sensitive to them.

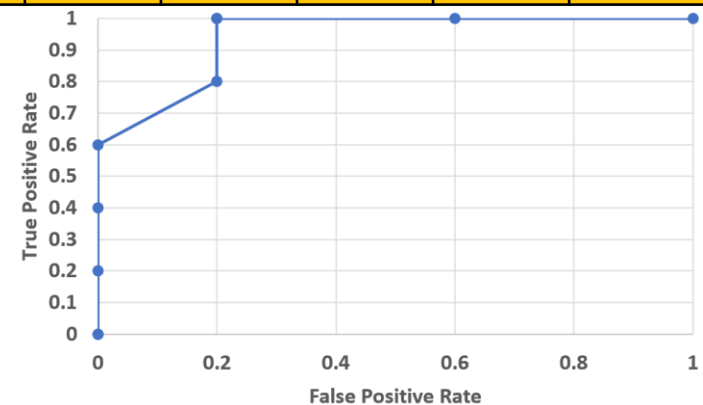
The Thresholding Problem in Classification

- Pinocchio's nose
 - Assume that liars have longer noses (wooden dummies only! 😊)
 - We need to set a threshold on the nose length (input feature) above which we classify the subject as a liar (label = yes).
 - The same principle applies to classification of:
 - A growth as a tumor based on size
 - Blood pressure levels as hypertension
 - An email as SPAM/Not-SPAM, Misinfo/Not-Misinfo based on probability scores
 - A search item as match/not-match based on similarity scores, e.g.
 - spoken term detection
 - keyword spotting
 - voice biometrics
- So, the question is where to place the cutoff
 - Can we exhaustively try all cutoffs over a bounded score?
 - Yes, but what do we track?
 - Precision/Recall?
 - False acceptances/False rejections?
 - True positives/False positives?
- Say hello to RoC curves!

Receiver Operating Characteristic (RoC)

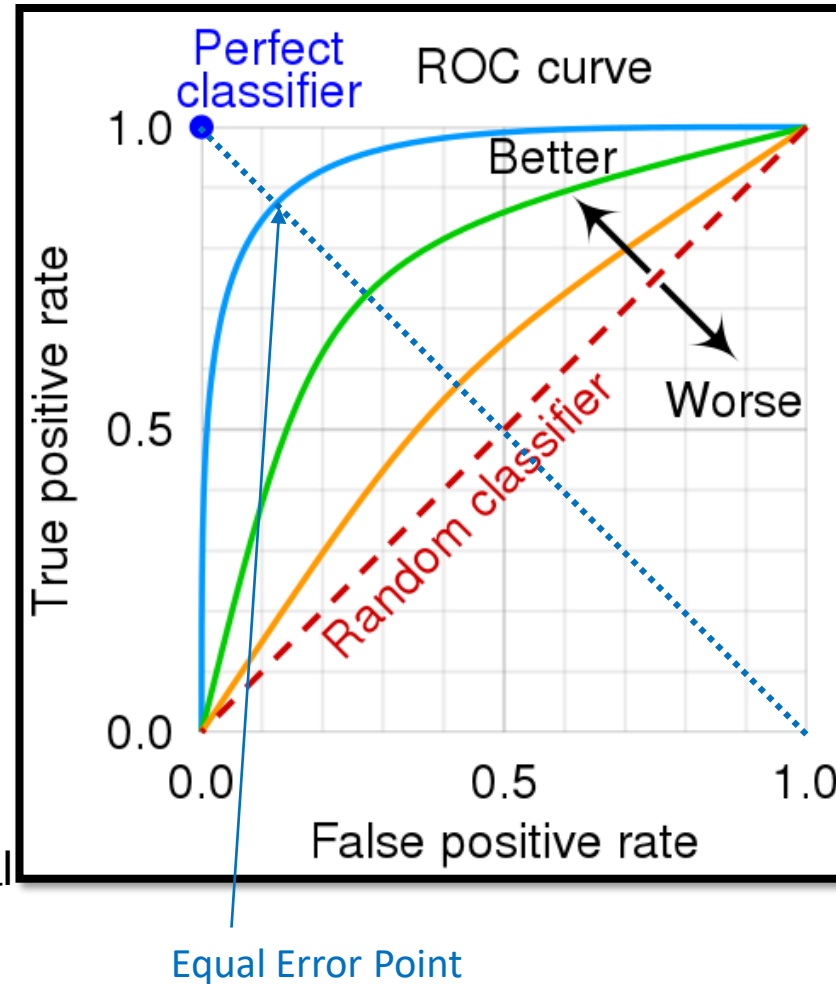
- A graphical plot that illustrates the diagnostic ability of a binary classifier as its discrimination threshold is varied
- The method was originally developed for operators of military radar receivers starting in 1941, which led to its name.
- Plot the *true positive rate* (TPR) – sensitivity – against the *false positive rate* (FPR) – (1 - specificity) at various threshold settings

Height h (inches)	Output Score (probability)	Adult (gold)	Thresholds																			
			0.1		0.2		0.3		0.4		0.5		0.6		0.7		0.8		0.9		1.0	
12	0.12	n	y	fp	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn
82	0.82	y	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	n	fn	n	fn
18	0.18	n	y	fp	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn
60	0.6	y	y	tp	y	tp	y	tp	y	tp	y	tp	n	fn	n	fn	n	fn	n	fn	n	fn
72	0.72	y	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	n	fn	n	fn	n	fn
55	0.55	n	y	fp	y	fp	y	fp	y	fp	y	fp	n	tn	n	tn	n	tn	n	tn	n	tn
48	0.48	y	y	tp	y	tp	y	tp	y	tp	n	fn	n	fn	n	fn	n	fn	n	fn	n	fn
24	0.24	n	y	fp	y	fp	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn
26	0.26	n	y	fp	y	fp	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn
68	0.68	y	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	n	fn	n	fn	n	fn	n	fn
tp			5		5		5		5		4		3		2		1		0		0	
fn			0		0		0		0		1		2		3		4		5		5	
fp			5		3		1		1		1		0		0		0		0		0	
tn			0		2		4		4		4		5		5		5		5		5	
1-Specificity (FPR)			1		0.6		0.2		0.2		0.2		0		0		0		0		0	
Sensitivity (Recall, TPR)			1		1		1		1		0.8		0.6		0.4		0.2		0		0	
Precision			0.5		0.625		0.8333		0.8333		0.8		1		1		1		NAN		NAN	



Characteristics

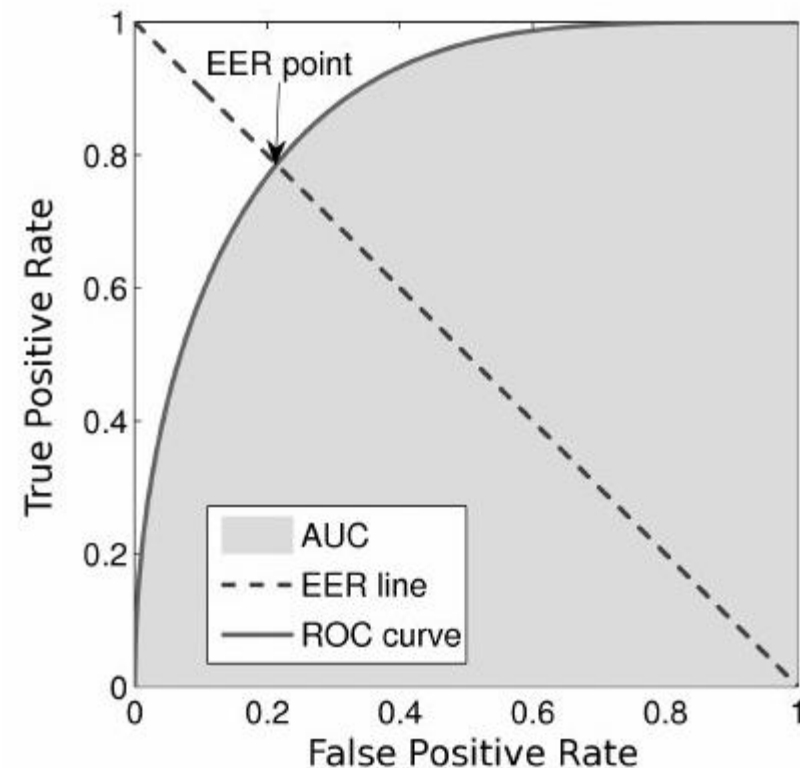
- The best possible prediction method would yield a point in the upper left corner (0,1) i.e. 100% sensitivity (no false negatives) and 100% specificity (no false positives)
- A random guess would give a point along a diagonal line (the line of no-discrimination) from the left bottom to the top right corners (TPR = FPR)
- The **red diagonal** divides the ROC space.
 - Points above the diagonal represent good classification (better than random)
 - Points below the line represent bad results (worse than random)
 - The output of a consistently bad predictor could simply be inverted to obtain a good predictor.
- The blue diagonal is the Equal Error Diagonal
 - Here $FPR = FNR$
 - Where $FNR = 1 - TPR$
 - A viable way to locate desired threshold
 - Smaller is better (in the graph: higher and to the left)



https://en.wikipedia.org/wiki/Receiver_operating_characteristic

More Characteristics

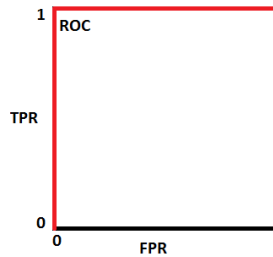
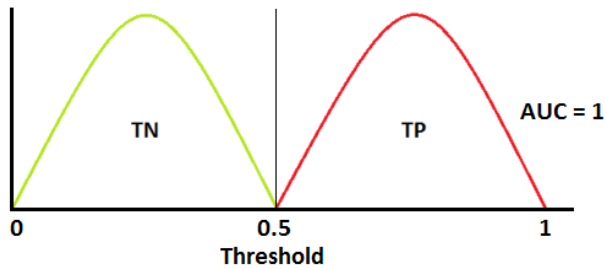
- It is hard to compare classifiers using ROC curves
- A way around that is to use AUC – Area under the ROC Curve, A' (pronounced "a-prime") or "c-statistic" ("concordance statistic").
 - Larger is better
- “AUC ROC can be interpreted as the probability that the scores given by a classifier will rank a randomly chosen positive instance higher than a randomly chosen negative one.” (Page 54, Learning from Imbalanced Data Sets, 2018)
- For imbalanced datasets: “ROC analysis does not have any bias toward models that perform well on the minority class at the expense of the majority class—a property that is quite attractive when dealing with imbalanced data.” (Page 27, Imbalanced Learning: Foundations, Algorithms, and Applications, 2013)



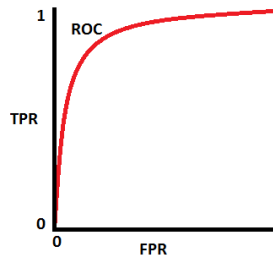
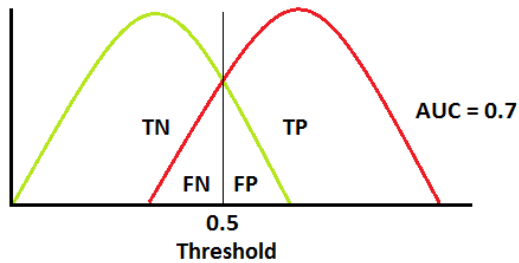
Tronci, Roberto, Giorgio Giacinto, and Fabio Roli. "Dynamic score combination: A supervised and unsupervised score combination method." In *International Workshop on Machine Learning and Data Mining in Pattern Recognition*, pp. 163-177. Springer, Berlin, Heidelberg, 2009.

https://www.researchgate.net/publication/225180361_Dynamic_Score_Combination_A_Supervised_and_Unsupervised_Score_Combination_Method/figures?lo=1

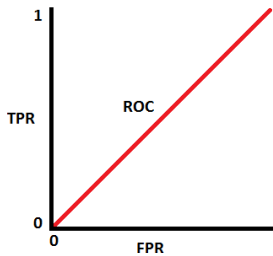
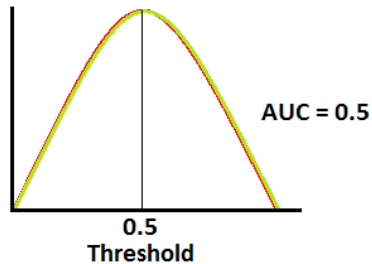
Examples



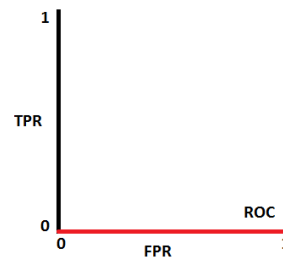
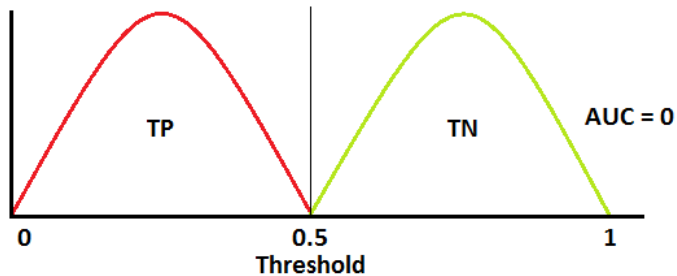
$$AUC \rightarrow 1$$



$$AUC \approx 0.7$$



$$AUC \approx 0.5$$

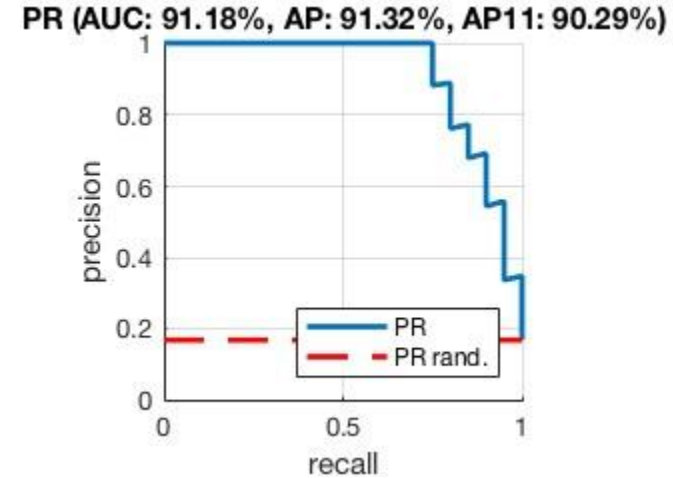


$$AUC \rightarrow 0$$

<https://towardsdatascience.com/understanding-auc-roc-curve-68b2303cc9c5>

Precision-Recall Curve

- The precision-recall curve shows the tradeoff between precision and recall for different thresholds. A high area under the curve represents both high recall and high precision
- High scores for both show that the classifier is returning accurate results (high precision), as well as returning a majority of all positive results (high recall).



<https://www.vlfeat.org/overview/plots-rank.html>, https://scikit-learn.org/stable/auto_examples/model_selection/plot_precision_recall.html

Discussion: PR vs ROC

Assuming a "positive" class 1 and a "negative" class 0. $\hat{y}$ is our estimate of the true class label y .

$$\text{Precision} = P(y = 1 | \hat{y} = 1)$$

$$\text{Recall} = \text{Sensitivity} = P(\hat{y} = 1 | y = 1)$$

$$\text{Specificity} = P(\hat{y} = 0 | y = 0)$$

- $P(y = 1)$ is the baseline probability depending on how common the event is in the real world
- $P(\hat{y} = 1)$ is the probability that our classifier will classify a sample as positive
- ROC curves will be the same regardless of $P(y = 1)$
- PR curves may be more useful in practice for needle-in-haystack type problems or problems where the "positive" class is more interesting than the negative class.

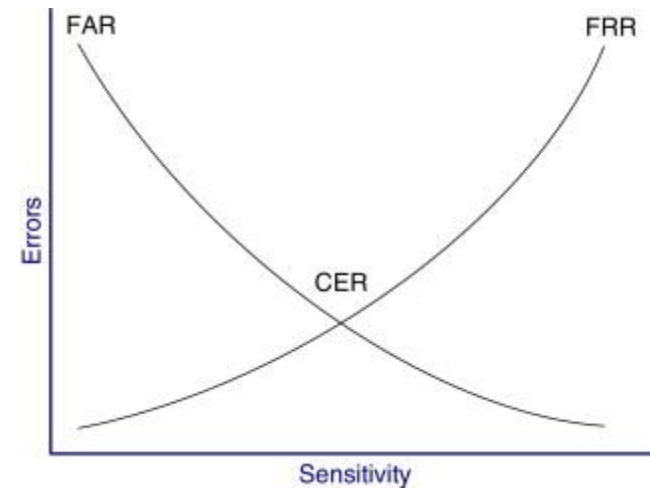
Bottomline:

- Use precision and recall to focus on a small positive class
 - When $P(y = 1)$ is low and the ability to detect correctly positive samples is our main focus (correct detection of negatives examples is less important to the problem).
- Use ROC when the detection of both classes is equally important
 - When we want to give equal weight to the prediction ability of both classes.
- Use ROC when the positives are the majority or switch the labels and use precision and recall
 - When the positive class is larger, use the ROC metrics because the precision and recall would reflect mostly the ability of prediction of the positive class and not the negative class which will naturally be harder to detect due to the smaller number of samples.
 - If the negative class (the minority in this case) is more important, we can switch the labels and use precision and recall.

<https://stats.stackexchange.com/questions/7207/roc-vs-precision-and-recall-curves>, <https://towardsdatascience.com/what-metrics-should-we-use-on-imbalanced-data-set-precision-recall-roc-e2e79252aeba>

A word on Biometric Systems

- The crossover error rate describes the point where the false reject rate (FRR) and false accept rate (FAR) are equal. CER is also known as the equal error rate (EER). The crossover error rate describes the overall accuracy of a biometric system.
- As the **sensitivity** of a biometric system increases, FRRs will rise and FARs will drop. Conversely, as the sensitivity is lowered, FRRs will drop and FARs will rise.
- CER is better when lower.
- Authentication algorithms need to simultaneously minimize the *permeability* to intruders; therefore, they must be *demanding*, and to maximize the *comfort* level, therefore, to be *permissive*.
 - This contradiction is the base for the optimization problem in authentication algorithms and the measure of success for the overall precision of an algorithm and of its usability is the CER.



This is not recall!
This is the calibration sensitivity of the biometric device.

For more details please visit
<http://aghaaliraza.com>



Thank you!